Ex No: 2

16/7/2026

Study Of Different Types Of Network Cables

Aim:

To study and understand different types of network cable, to make your own ethernet cross over cable / straight cable.

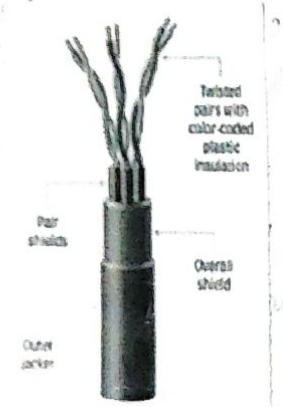
Different Types Of Network Cable:

1. Unshielded Twisted pair cable
2. Shielded Twisted pair cable
3. Coaxial cable
4. Fibre optic cable.

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Applications
UTP	Category 3	10 Mbps	Advantages: -> Cheapest in cost -> Easy to install as they have small overall diameter Disadvantages: -> More prone to EMI	10 Base-T ethernet
	Category 5	Upto 100 mbps		Fast Ethernet, Gigabit Ethernet
	Category 5e	1 Gbps		
STP	Category 6, 6a	10 Gbps	Advantages: -> Shielded -> Faster than UTP -> Less prone to noise	-> Gigabit ethernet -> widely used in data centers
SSTP	Category 7	10 Gbps	Disadvantages: -> Expensive -> Greater installation process	-> Gigabit ethernet, 10 G Ethernet

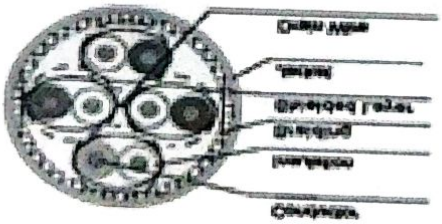


- UTP



Shielded twisted pair

- STP





- SSTP

Coaxial Cable	RG-6 RG-59 RG-11	10 - 100 Mbps	Advantages: -> High bandwidth -> Immune to interference -> Versatile Disadvantages: -> Cost -> Size is bulky	-> Speed of signal is 500m -> High speed internet connection
Fibre Optics Cable	Singlemode Multimode	100Gbps	Advantages: -> High speed -> High bandwidth -> High security Disadvantages: -> Expensive	-> Maximum distance of fibre optics cable is around 100 m

Make your own Ethernet Cross-Over Cable / Straight Cable.

Tools and parts needed:

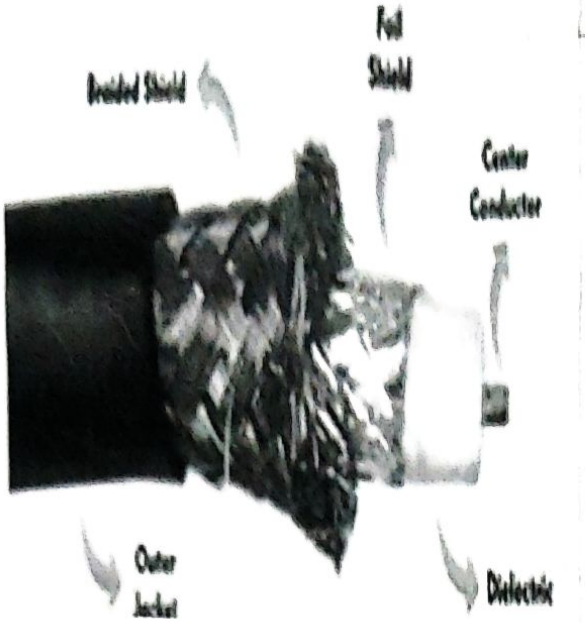
-> Ethernet cabling. Cat5e is certified for gigabit support, but CAT5 cabling works as well

-> A crimping tool

-> Two RJ45 plugs

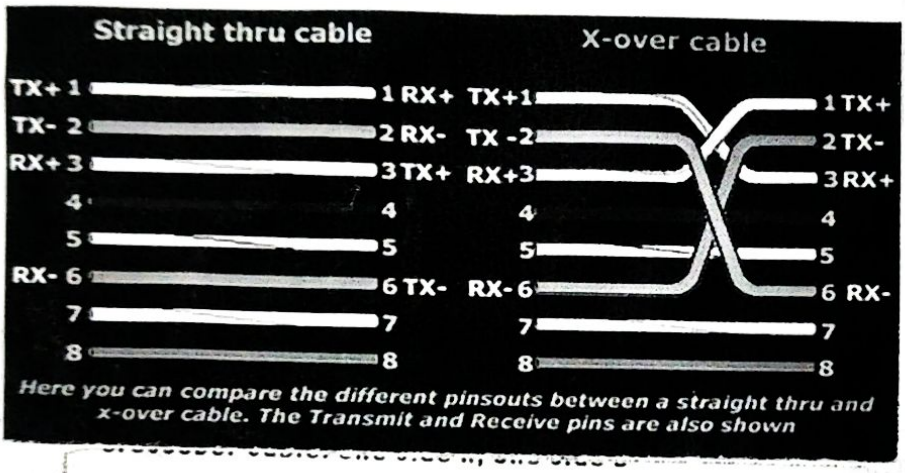
-> Optional two plug shields.

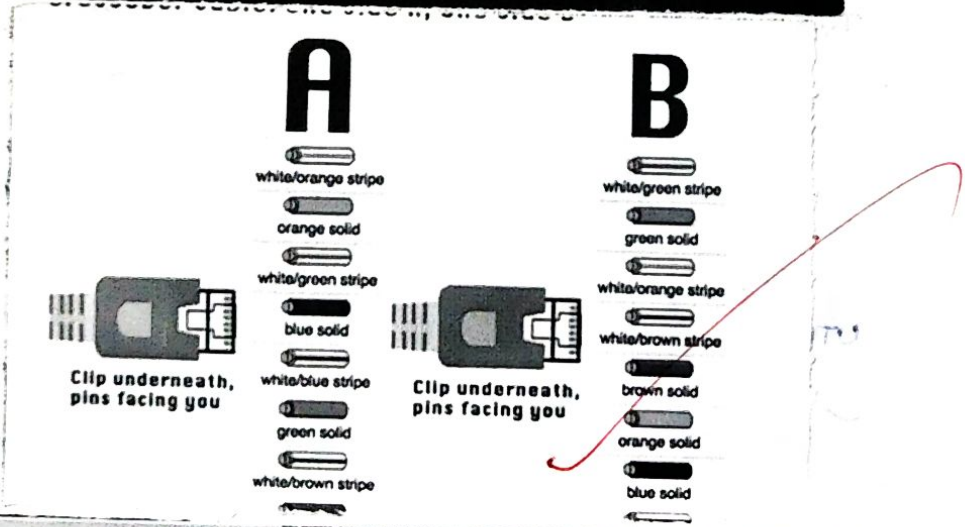
Coaxial cable

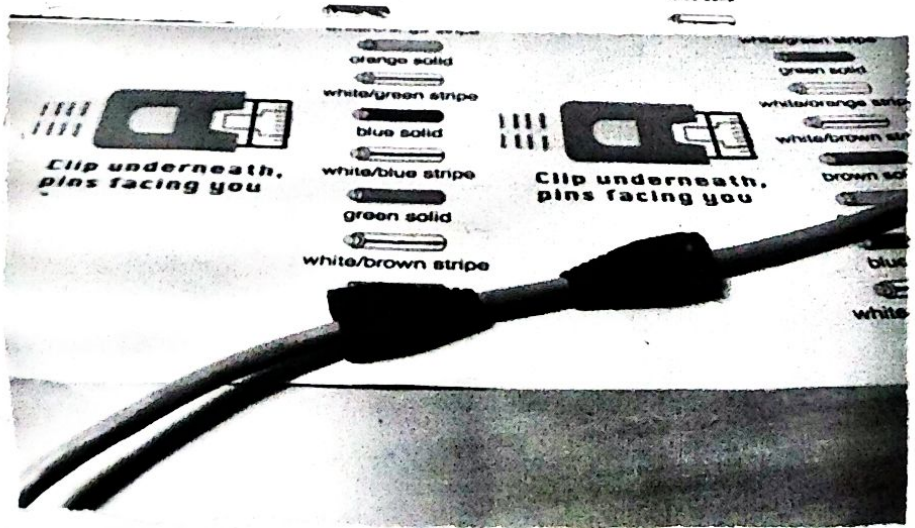




Fibre Optics Cable







STEP 1:

To start construction of the device, begin by threading shields on to the cable

STEP 2:

Next, strip approximately 1.5cm of cable shielding from both ends.

STEP 3:

After you will need to untangle the wires, there should be four twisted pairs.

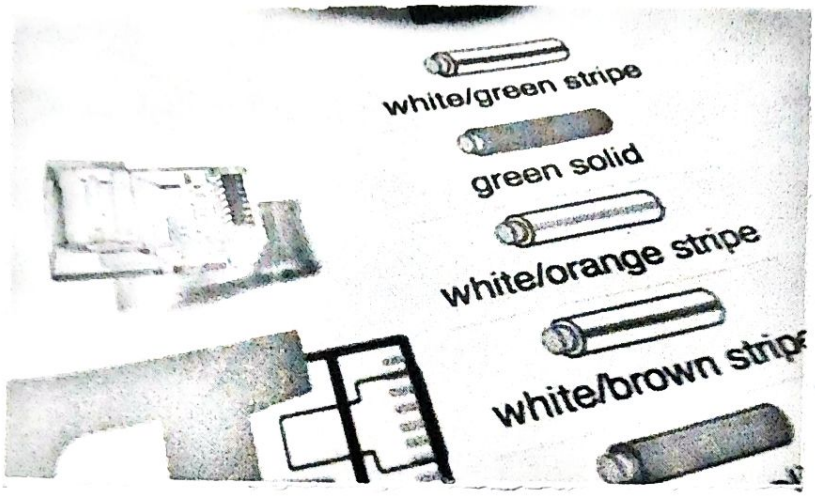
STEP 4:

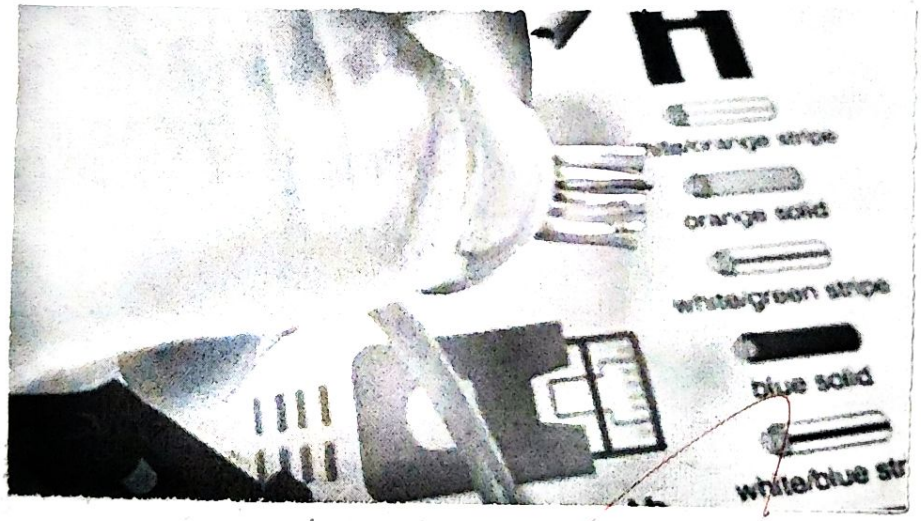
Once the order is correct, bunch them to gather in a line and if there are any that stick out further than others snip them back to create an even level. the difficult aspect is placing the RJ45 plug without messing up the order.

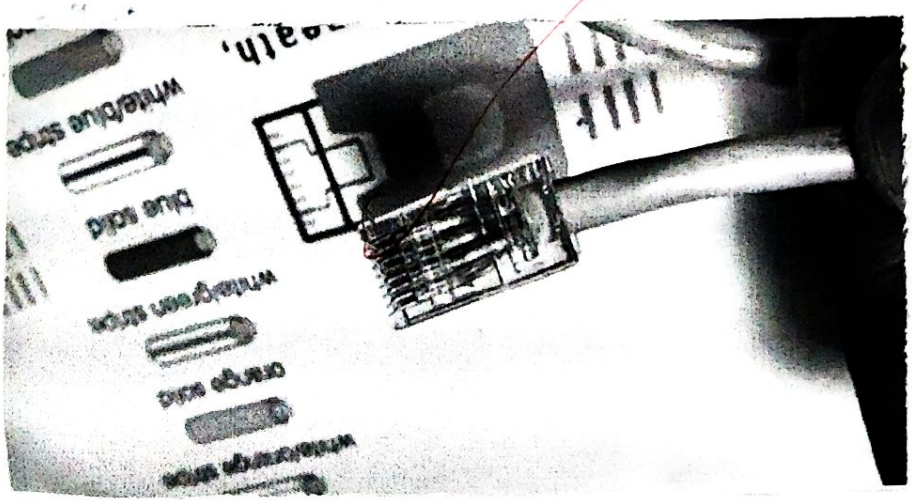
STEP 5:

Next push the cable right in. The notch at the end to be just over the cable shielding and if it isn't, that means that you stripped off too much shielding


A
B


white/green stripe
green solid
white/orange stripe
white/brown stripe


orange solid
white/green stripe
blue solid



STEP 6:

After the wires are securely getting inside the plug insert it into the crimping tool and push down

STEP 7:

Lastly repeat for the other end using diagram B.

To test it, plug in and attempt to connect two devices directly.

23/7/24.

RESULT:

Thus, different types of network cables were studied successfully and straight through and cross over Ethernet cables were prepared and tested successfully.